

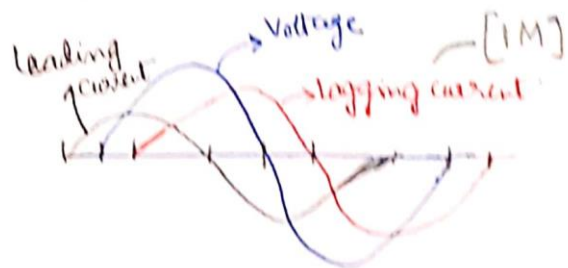
1A:

(a) In a circuit with alternating current, the value of voltage & current vary sinusoidally.

- Lagging Current is the current that reaches 'maximum value' upto 90° later than the voltage that produces it. Alternatively, the lagging voltage is the voltage that reaches maximum value upto 90° later than the current within it. [1M]

- Vice-versa is true for 'leading' term.

wave form:



(b) Apparent Power: Apparent power is the product of the voltage & current, flowing instantaneously through an electrical element. It is the total power flowing [0.5M] instantaneously.

Complex Power: Complex Power is the vector sum of Real & Reactive Power. It is associated with finite magnitude & phase angle. [0.5M]

Difference — Complex Power ^{signifies} is the amount of magnitude of Real & reactive power consumed by an element, while [1M] voltage & current across that must be synchronized.

However, Apparent Power is when volt & current are synchronized/observed in same time frame.

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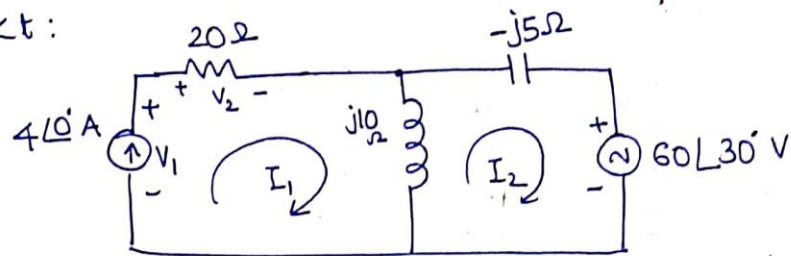
- Frequency is the number of complete cycles that the wave completes in a given amount of time, usually measured in Hz. $\text{Freq.} = 1/(\text{Time Period in sec.})$ [0.5 M]

The Sharp-ness of resonance obtained in a RLC circuit, is determined by Quality Factor. It basically signifies the sharpness of resonance peak, & the amplification of the voltages/ currents in series/parallel resonating circuits.

If Q value is high, resonance is sharp. Low values of Q-factor implies weak resonance [1.5 M]

1B

Given CKT:



apply mesh-analysis:

Mesh-1: $I_1 = 4 \angle 0^\circ \text{ A}$

Mesh-2: $(j10 - j5) I_2 - j10 I_1 + 60 \angle 30^\circ = 0$

(OR) $-j10(I_2 - I_1) - (-j5 I_2) - 60 \angle 30^\circ = 0$

$\Rightarrow j5 I_2 = -60 \angle 30^\circ + j40 \Rightarrow I_2 = 10.58 \angle 79.1^\circ \text{ A}$

For voltage source, current flowing is $I_2 = 10.58 \angle 79.1^\circ \text{ A}$
& voltage across it is $60 \angle 30^\circ$.

$\therefore \text{Avg. Power} = 60 \times 10.58 \cos(30 - 79.1)$
 $= \underline{415.6 \text{ W}}$

$\therefore$ Direction of I_2 is opposite; the power is absorbed by voltage source.

$\therefore \text{Avg. Power Supplied by volt. source} = -415.6 \text{ W}$

Current through current source is $I_1 = 4 \angle 0^\circ \text{ A}$; voltage across it is,

$$\begin{aligned} V_1 &= 20I_1 + j10(I_1 - I_2) \\ &= 80 + j10(4 - 2 - j10 \cdot 39) \\ &= 184.98 \angle 6.21^\circ \text{ V.} \end{aligned} \quad \text{---[IM]}$$

$\therefore$ Avg. Power supplied by current source is,

$$+ (184.98) \times (4) \times \cos(6.21 - 0) = + \underline{735.6 \text{ W}} \quad \text{---[IM]}$$

Since, there is only one ^{passive} element consuming avg. power (resistor);

$$\begin{aligned} \text{the Avg. Power absorbed by resistor} &= 735.6 - 415.6 \text{ W} \\ &= \underline{320 \text{ W}} \quad \text{---[IM]} \end{aligned}$$

Alternatively,

$$\begin{aligned} \text{Avg. Power absorbed by resistor} &= I_{\text{rms}} \cdot \overbrace{8(20I_{\text{rms}})}^{\text{volt across res}} \cos 0 \\ &= 4 \times 80 = \underline{320 \text{ W}} \end{aligned}$$

$$\nabla \text{ Avg. Power absorbed by Capacitor} = \underline{0 \text{ W}} \quad \text{---[IM]}$$

($\because$ Capacitor only accounts for reactive power)

(a) For Coil-A, $|Z_1| = \frac{V}{I} = \frac{120}{6} = 20 \Omega$.

$$P = R_1 \times I^2$$

$$\therefore R_1 = \frac{P}{I^2} = \frac{160}{36} = \underline{4.44 \Omega} \quad [1M]$$

$$\therefore X_1 = \sqrt{|Z_1|^2 - R_1^2} = \sqrt{20^2 - 4.44^2} = \underline{19.5 \Omega} \quad [1M]$$

For Coil-B, $|Z_2| = \frac{V}{I} = \frac{120}{12} = 10 \Omega$

$$R_2 = \frac{P}{I^2} = \frac{420}{10^2} = \underline{4.2 \Omega} \quad [1M]$$

$$\therefore X_2 = \sqrt{|Z_2|^2 - R_2^2} = \underline{9.075 \Omega} \quad [1M]$$

(b) When two coils are connected in series,

$$\begin{aligned} \bar{Z}_{eq} &= (R_1 + jX_1) + (R_2 + jX_2) \\ &= (R_1 + R_2) + j(X_1 + X_2) = \underline{8.64} + j\underline{28.575} \quad [1M] \\ &= \underline{29.85} \angle \underline{73.47^\circ} \end{aligned}$$

Thus, Current phasor when 100V is applied, $\bar{I} = \frac{\bar{V}}{\bar{Z}_{eq}} = \frac{100 \angle 0^\circ}{29.85 \angle 73.47^\circ}$
 $\uparrow$
 ref. phasor $[2M] = \underline{3.35} \angle \underline{-73.47^\circ}$

Resultant power factor = $\cos(-73.47^\circ) = \underline{0.295} \text{ Lagging} \quad [1M]$

1D.

At series resonance, ckt Max. Current (I_m) is given as,

$$I_m = \frac{V}{R} \quad \text{--- (1)}$$

at ^{any} $f (= \frac{\omega}{2\pi})$ frequency, the circuit impedance is given by,

$$Z = \sqrt{R^2 + (X_L - X_C)^2} \quad (\because \text{series RLC ckt})$$

$$\text{Circuit current, } I = \frac{V}{Z} \quad \text{--- (2)}$$

$$\text{From (1) \& (2), } \therefore \frac{I_m}{I} = \left(\frac{V}{R}\right) \cdot \frac{R}{Z} / \left(\frac{V}{Z}\right) = \frac{Z}{R}$$

$$\text{Given } \frac{I_m}{I} = 2; \Rightarrow 2 = \frac{Z}{R} \quad \text{--- [1M]}$$

$$\Rightarrow Z^2 = 4R^2$$

$$\Rightarrow R^2 + (X_L - X_C)^2 = 4R^2 \Rightarrow (X_L - X_C)^2 = 3R^2$$

$$\therefore X_L - X_C = \omega L - \frac{1}{\omega C} = \pm \sqrt{3}R$$

$$\swarrow \Rightarrow \omega L - \frac{1}{\omega C} = +\sqrt{3}R \quad \searrow \Rightarrow \omega L - \frac{1}{\omega C} = -\sqrt{3}R$$

$$\omega L - \frac{1}{\omega C} = +\sqrt{3}R$$

↓ solve

$$\omega_1 = \frac{\sqrt{3}R}{2L} + \sqrt{\frac{3}{4}\left(\frac{R}{L}\right)^2 + \frac{1}{LC}}$$

↓ I_m

rad/s
(Discard -ve 'w') [2M]
↓ I_m

$$\omega L - \frac{1}{\omega C} = -\sqrt{3}R$$

↓ solve

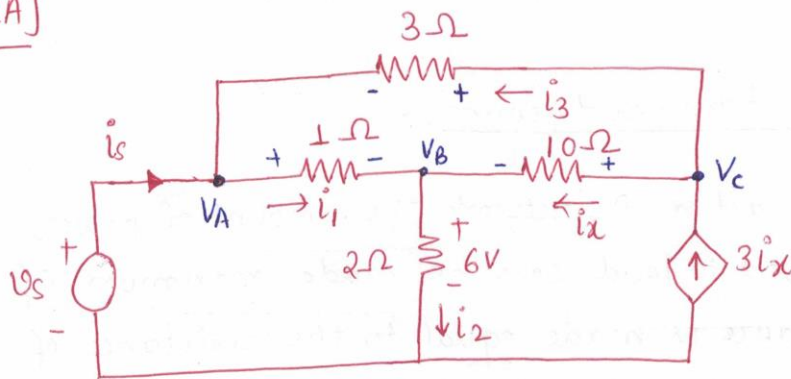
$$\omega_2 = -\frac{\sqrt{3}R}{2L} + \sqrt{\frac{3}{4}\left(\frac{R}{L}\right)^2 + \frac{1}{LC}}$$

↓ I_m

rad/s.

(Discard -ve 'w') [2M]
↓ I_m

Q.2A]



Solⁿ:- $i_2 = \frac{6V}{2\Omega} = 3A$. — [1M]

using KCL at node V_C .

$$3i_x = i_x + i_3$$

$$\therefore i_3 = 2i_x$$

At node V_A , $i_s + i_3 = i_1$

at node V_B , $i_1 + i_x = 3$

using KVL in upper loop,

$$-3i_3 - i_1 + 10i_x = 0$$

$$-3 \times (2i_x) - i_1 + 10i_x = 0$$

$$i_1 = 4i_x$$

$$4i_x + i_x = 3 \quad \therefore i_x = 0.6A \quad \text{--- [1M]}$$

$$\therefore i_1 = 3 - i_x = 2.4A \quad \text{[1M]}$$

$$i_3 = 2i_x = 1.2A \quad \text{[1M]}$$

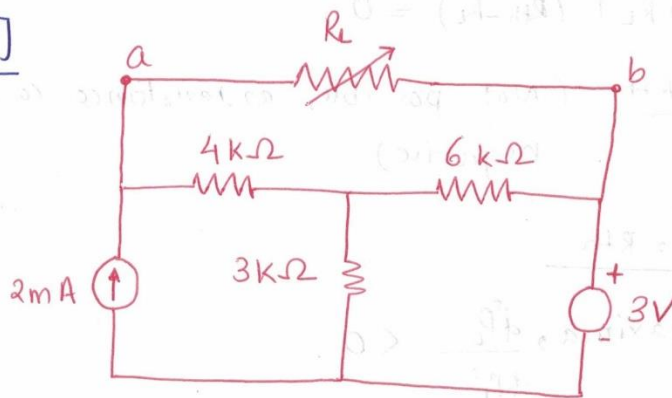
$$\text{[0.5M]} \quad i_s = i_1 - i_3 = 1.2A \quad \text{[1M]}$$

using KVL, $v_s - i_1 - 6V = 0 \quad \therefore v_s = 8.4V \quad \text{[0.5]}$

$$\frac{v_s}{i_s} = \frac{8.4}{1.2} = 7.0$$

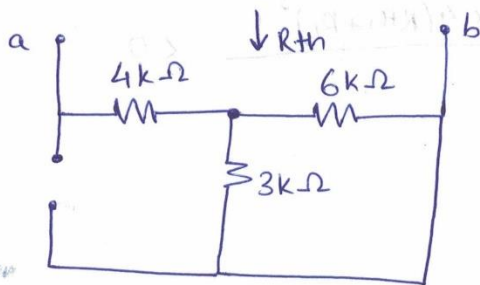
[1 M for KVL]

Q.2B]



* Find the power delivered to load under maximum power transfer condⁿ.

Solⁿ:- for finding R_{th} -



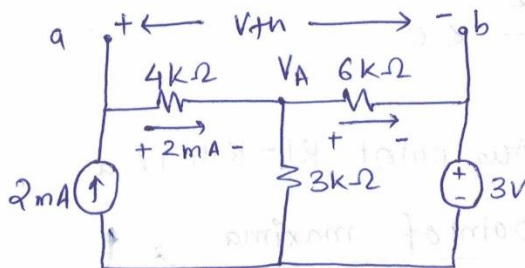
$$R_{th} = 4k\Omega + 3k\Omega \parallel 6k\Omega$$

$$= 4k\Omega + \frac{3 \times 6}{3+6} k\Omega$$

$$R_{th} = 6k\Omega$$

[2M]

for finding V_{th} -



Using Nodal Analysis -

$$2mA = \frac{V_A}{3k} + \frac{V_A - 3}{6k}$$

$$2 \times 6 = 2V_A + V_A - 3$$

$$V_A = 5V$$

[2M]

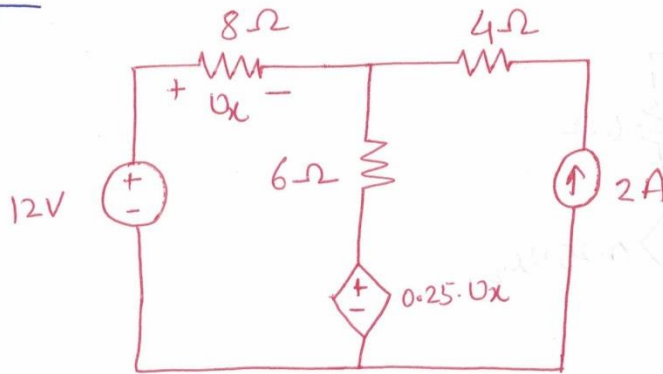
$$i_{6k} = \frac{V_A - 3}{6} = \frac{2}{6} = \frac{1}{3} mA$$

using KVL, $V_{th} - 8 - 6 \times \frac{1}{3} = 0 \therefore V_{th} = 10V$.

$$Power = \frac{V_{th}^2}{4 \times R_{th}} = \frac{10^2}{4 \times 6 \times 10^3} = 4.166 mW$$

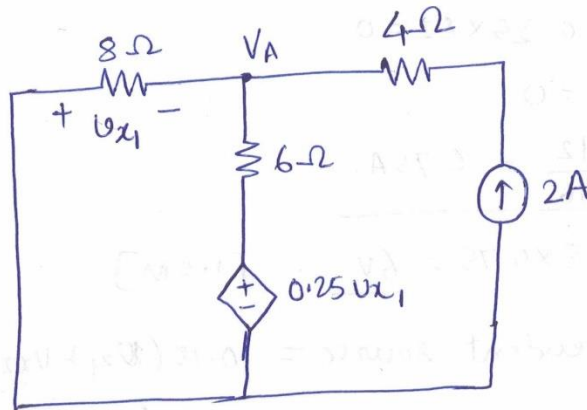
[1M]

Q.2c)



Solⁿ:-

a) Considering 2A source alone -



using KCL at node V_A -

$$2 = \frac{V_A}{8} + \frac{V_A - 0.25V_{x1}}{6}$$

by inspection, $V_A = -V_{x1}$

$$2 = \frac{-V_{x1}}{8} + \frac{-V_{x1} - 0.25V_{x1}}{6}$$

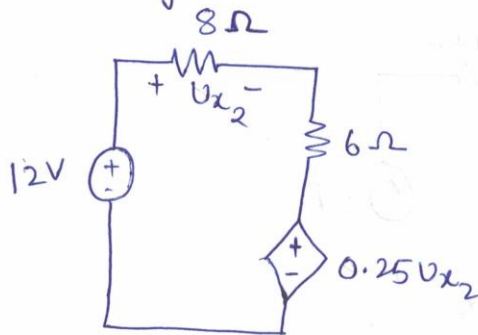
$$2 = \frac{-3V_{x1} - 5V_{x1}}{24}$$

$$-V_{x1} = 6V$$

$$V_{x1} = -6V$$

[1.5M]

Considering 12V source alone -



$$12 - 8I - 6I - 0.25V_{x2} = 0$$

$$V_{x2} = 8 \times I$$

$$\therefore 12 - 14I - 0.25 \times 8I = 0$$

$$12 - 16I = 0$$

$$I = \frac{12}{16} = 0.75A$$

$$V_{x2} = 8 \times 0.75 = 6V. \quad [1.5M]$$

$$\begin{aligned} \text{Magnitude of dependent source} &= 0.25(V_{x1} + V_{x2}) \\ &= 0.25(-6 + 6) = 0V. \quad [1M] \end{aligned}$$

$$\begin{aligned} \text{b) for 2A alone, } V_A &= 6V, \quad -V_A = -6V, \quad i_1 = \frac{V_A - 0.25V_{x1}}{6} \\ &= \frac{6 + 1.5}{6} = \frac{7.5}{6} = 1.25A \end{aligned}$$

$$\text{for } 12 \text{ alone } I = 0.75A$$

$$\therefore I_{6\Omega} = 1.25A + 0.75 = 2A. \quad [2M]$$

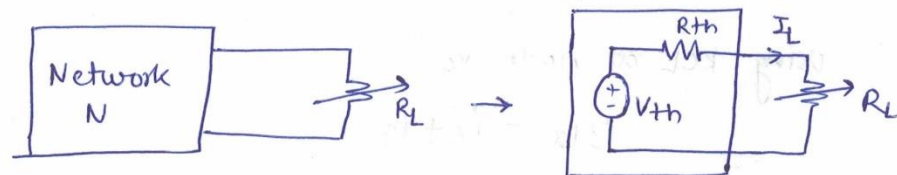
$$\text{c) Power delivered or absorbed} = 2 \times 0.25V_x = 0W. \quad [1M]$$

Q. 2D)

a) Maximum Power transfer theorem:-

In any linear active DC network, the amount of power to be transferred to load can be made maximum if the load resistance is made equal to the resistance of the network seen from the load terminals. [2M]

Proof:-



Ther. equivalent of network.

The network, N can be equivalently represented using Thevenin's network containing V_{th} & R_{th} as shown

$$\therefore \text{Load current } I_L = \frac{V_{th}}{R_{th} + R_L}$$

$$\text{Power drawn by load} = I_L^2 R_L$$

$$P_L = \frac{V_{th}^2}{(R_{th} + R_L)^2} \times R_L$$

$$\text{for max. power } \frac{dP_L}{dR_L} = 0 \rightarrow V_{th}^2 \left[\frac{(R_{th} + R_L)^2 - R_L \cdot 2(R_{th} + R_L)}{(R_{th} + R_L)^4} \right] = 0$$

$$\frac{V_{th}^2 [R_{th}^2 + \cancel{2R_L R_{th}} + R_L^2 - \cancel{2R_L R_{th}} - 2R_L^2]}{(R_{th} + R_L)^4} = 0$$

$$(R_{th}^2 - R_L^2) = 0$$

$$(R_{th} + R_L)(R_{th} - R_L) = 0$$

$R_L = -R_{th}$ (Not possible, as resistance can't be negative).

$$\therefore \underline{R_L = R_{th}}$$

for point of maxima, $\frac{d^2 P_L}{dR_L^2} < 0$.

$$\therefore \frac{d}{dR_L} \left[\frac{V_{th}^2 (R_{th}^2 - R_L^2)}{(R_{th} + R_L)^4} \right] < 0$$

$$V_{th}^2 \times \frac{(R_{th} + R_L)^4 (-2R_L) - (R_{th}^2 - R_L^2)^2 \times 4(R_{th} + R_L)^3}{(R_{th} + R_L)^8} < 0$$

Substituting $R_{th} = R_L$

$$\frac{V_{th}^2 \times (2R_L)^4 \times (-2R_L)}{(2R_L)^8} < 0$$

$$\frac{-2R_L V_{th}^2}{(2R_L)^3} < 0$$

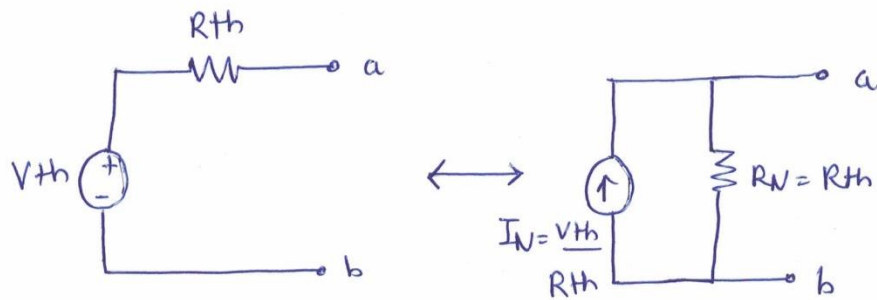
Since, $\frac{-V_{th}^2}{(2R_L)^3} < 0$, the point $R_L = R_{th}$ is a point of maxima.

Hence, Maximum power gets transferred to load when $R_L = R_{th}$.

[2M]

Q. 2D)

b)



For the two circuits to be called as equivalent -

i) open circuit voltage (i.e. voltage across a-b) must be same.

for Thev. equiv. circuit, $V_{ab} = V_{oc} = V_{th}$. $[I=0]$.

for Norton's equiv. circuit, $V_{ab} = V_{oc} = R_N \cdot I_N$

$$\therefore \underline{V_{th} = R_N \cdot I_N} \quad - (1) \quad [1M]$$

ii) Also, short circuit current must be same for both representations.

$$I_{sc} = \frac{V_{th}}{R_{th}} \quad \text{for Thev. equiv.}$$

$$I_{sc} = I_N \quad \text{for Norton's equiv.}$$

$$\frac{V_{th}}{R_{th}} = I_N \quad - (2)$$

Solving (1) & (2), we get, $\boxed{R_N = R_{th}}$ [1M]

$$\therefore I_N = \frac{V_{th}}{R_{th}}$$

Q. 3A

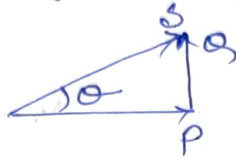
$$W_1 = 108.6 \text{ kW}$$

$$W_2 = 41.4 \text{ kW}$$

$$P = W_1 + W_2 = 108.6 + 41.4 = 150 \text{ kW} \quad [1]$$

$$Q = \sqrt{3} \times (W_1 - W_2) = \sqrt{3} \times (108.6 - 41.4) = 116.39 \text{ kVAR} \quad [1]$$

Since current lags the voltage, the circuit is inductive in nature.



$$\text{Complex power, } \bar{S} = P + jQ = 150 + j116.39 \text{ kVA} \quad [1]$$

$$\text{Apparent power, } S = \sqrt{P^2 + Q^2}$$

$$S = \sqrt{150^2 + 116.39^2} = 189.85 \text{ kVA} \quad [1]$$

$$\text{Power factor} = \frac{P}{S} = \frac{150}{189.85} = 0.79 \text{ lagging} \quad [1]$$

$$\text{If } \theta = 0^\circ, \quad W_1 = V_L I_L \cos(30 - \theta) = V_L I_L \cos(30^\circ)$$

$$W_2 = V_L I_L \cos(30 + \theta) = V_L I_L \cos(30^\circ)$$

$$\text{Hence } W_1 = W_2$$

$$\text{From previous case, } P = W_1 + W_2 = 150 \text{ kW}$$

$$\therefore W_1 = W_2 = \frac{P}{2} = 75 \text{ kW} \quad [2]$$

Q. 3B

Consider a balanced three-phase system. Assume the current lags the voltage by an angle ' θ '.

$$\text{Hence, if } v_a = \sqrt{2} V_p \sin(\omega t), \quad v_b = \sqrt{2} V_p \sin(\omega t - \frac{2\pi}{3}),$$

$$\& v_c = \sqrt{2} V_p \sin(\omega t - \frac{4\pi}{3}), \text{ then } i_a = \sqrt{2} I_p \sin(\omega t - \theta),$$

$$i_b = \sqrt{2} I_p \sin(\omega t - \frac{2\pi}{3} - \theta) \& i_c = \sqrt{2} I_p \sin(\omega t - \frac{4\pi}{3} - \theta).$$

Instantaneous power in each phase is given by:

$$P_a = v_a i_a, \quad P_b = v_b i_b, \& P_c = v_c i_c \quad [1]$$

Phase 'a' power

$$P_a = V_a i_a = \sqrt{2} V_p \sin(\omega t) \times \sqrt{2} I_p \sin(\omega t - \theta)$$

$$= 2 V_p I_p \sin(\omega t) \times \sin(\omega t - \theta)$$

$$= 2 V_p I_p \sin(\omega t) \times [\sin(\omega t) \cos \theta - \cos(\omega t) \sin \theta]$$

$$= 2 V_p I_p [\sin^2 \omega t \cos \theta - \sin \theta \cos \omega t \sin \omega t]$$

$$= 2 V_p I_p \left[\left(\frac{1 - \cos 2\omega t}{2} \right) \cos \theta - \sin \theta \frac{\sin 2\omega t}{2} \right]$$

$$P_a = V_p I_p \cos \theta (1 - \cos 2\omega t) - V_p I_p \sin \theta \sin 2\omega t \quad [1]$$

Similarly $P_b = V_p I_p \cos \theta \left[1 - \cos \left\{ 2\left(\omega t - \frac{2\pi}{3}\right) \right\} \right]$
 $- V_p I_p \sin \theta \sin \left\{ 2\left(\omega t - \frac{2\pi}{3}\right) \right\} \quad [0.5]$

& $P_c = V_p I_p \cos \theta \left[1 - \cos \left\{ 2\left(\omega t - \frac{4\pi}{3}\right) \right\} \right]$
 $- V_p I_p \sin \theta \sin \left\{ 2\left(\omega t - \frac{4\pi}{3}\right) \right\} \quad [0.5]$

Resultant power, $P = P_a + P_b + P_c$

$$\Rightarrow P = 3V_p I_p \cos \theta - V_p I_p \cos \theta \times [\cos(2\omega t) + \cos(2\omega t - \frac{4\pi}{3}) + \cos(2\omega t - \frac{8\pi}{3})]$$
$$- V_p I_p \sin \theta \times [\sin 2\omega t + \sin(2\omega t - \frac{4\pi}{3}) + \sin(2\omega t - \frac{8\pi}{3})]$$

The terms multiplied by $V_p I_p \cos \theta$ & $V_p I_p \sin \theta$ are zero as there are summation of 3 quantities which are 120° apart.

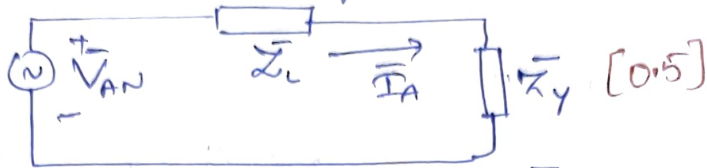
Hence $P = 3V_p I_p \cos \theta$ which is a constant.

[2]

Q.3C

Since $\bar{V}_{AB} = 340 \angle 0^\circ \text{ V}$, $\bar{V}_{BC} = 340 \angle -120^\circ$ & $\bar{V}_{CA} = 340 \angle -240^\circ \text{ V}$.
[0.5]

Solving the circuit for phase 'A'.



Per phase impedance $= \bar{Z}_L + \bar{Z}_Y = 2.5 \angle 30^\circ + 20 \angle 60^\circ \Omega$
 $\bar{Z} = 22.2 \angle 56.77^\circ \Omega$ [1]

Hence resultant power factor $= \cos 56.77^\circ$
 $= 0.548$ lagging [1]

Since $\bar{V}_{AB} = 340 \angle 0^\circ \text{ V}$

$\Rightarrow \bar{V}_{AN} = \frac{340}{\sqrt{3}} \angle -30^\circ \text{ V}$ [0.5]

$\therefore \bar{I}_A = \frac{\bar{V}_{AN}}{\bar{Z}} = \frac{\frac{340}{\sqrt{3}} \angle -30^\circ}{22.2 \angle 56.77^\circ} = 8.84 \angle -86.77^\circ$ [1] Ampere

$\therefore \bar{V}_{an} = \bar{I}_A \bar{Z}_Y = (8.84 \angle -86.77^\circ) \times (20 \angle 60^\circ) \text{ V}$
 $= 176.84 \angle -26.77^\circ \text{ V}$ [1]

$\therefore$ Magnitude of phase voltage at load terminal $= 176.8 \text{ V}$ [0.5]

& $\bar{V}_{ab} = \sqrt{3} |\bar{V}_{an}| \angle \theta_{\bar{V}_{an}} + 30^\circ = \sqrt{3} \times 176.8 \angle (-26.77 + 30)$
 $= 306.2 \angle 3.23^\circ \text{ V}$ [1]

Hence $\bar{V}_{bc} = 306.2 \angle (3.23 - 120) = 306.2 \angle -116.77^\circ \text{ V}$
[0.5] & $\bar{V}_{ca} = 306.2 \angle (-116.77 - 120) = 306.2 \angle -236.77^\circ \text{ V}$

$\therefore$ Magnitude of line-to-line voltage at load terminals is 306.2 V [0.5]

Current in other phases can be obtained using $\bar{I}_A$.

Since $\bar{I}_A = 8.84 \angle -86.77^\circ \text{ A}$, therefore

$\bar{I}_B = 8.84 \angle -206.77^\circ \text{ A}$, & $\bar{I}_C = 8.84 \angle -326.77^\circ \text{ A}$.
[0.5]

Power from source [1.5]

$$P_s = \sqrt{3} V_L I_L \cos \theta$$

$$P_s = \sqrt{3} \times 340 \times 8.84 \times 0.548$$

$$P_s = 2.852 \text{ kW}$$

$$Q_s = \sqrt{3} V_L I_L \sin \theta$$

$$Q_s = \sqrt{3} \times 340 \times 8.84 \times \sin(56.7^\circ)$$

$$Q_s = 4.354 \text{ kVAR}$$

Power delivered to load [1.5]

$$P_Y = 3 I^2 Z_Y \cos \theta_Y$$

$$P_Y = 3 \times 8.84^2 \times 20 \times \cos 60^\circ$$

$$P_Y = 2.344 \text{ kW}$$

$$Q_Y = 3 I^2 Z_Y \sin \theta_Y$$

$$Q_Y = 3 \times 8.84^2 \times 20 \times \sin 60^\circ$$

$$Q_Y = 4.06 \text{ kVAR}$$

Power lost in the line [1.5]

$$P_L = 3 I^2 Z_L \cos \theta_L$$

$$P_L = 3 \times 8.84^2 \times 2.5 \times \cos 30^\circ$$

$$P_L = 0.507 \text{ kW}$$

$$Q_L = 3 I^2 Z_L \sin \theta_L$$

$$Q_L = 3 \times 8.84^2 \times 2.5 \times \sin(30^\circ)$$

$$Q_L = 0.293 \text{ kVAR}$$

A) Induction Motor is of 4 Pole.
 supplied frequency $f = 50 \text{ Hz}$.
 Rotor emf frequency $f_r = 2 \text{ Hz}$

a) slip $s = \frac{f_r}{f} = \frac{2}{50} = 0.04$

$\therefore$ slip in percentage is 4% (2)

b) Rotor speed = N_r

$s = \frac{N_s - N_r}{N_s}$ [N_s = Synchronous speed]

Here $N_s = \frac{120 \times f}{P} = \frac{120 \times 50}{4} = 1500 \text{ rpm}$

$s = 0.04$

$\therefore N_r = 1500 - 0.04 \times 1500 = 1440 \text{ rpm}$ (2)

$\Rightarrow$ Synchronous Speed $N_s = \frac{120 \times f}{P} = 1500 \text{ rpm}$ (2)

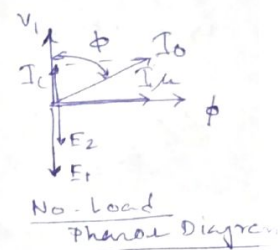
B) Full Load 'VA' $S = 5 \times 10^3 \text{ VA}$

Primary voltage $V_p = 230 \text{ V}$

Secondary voltage $V_s = 115 \text{ V}$

Core Loss $P_c = 100 \text{ W}$

No load VA $S_{\text{no-load}} = 400 \text{ VA}$



a) Iron Loss component of current

$I_c = I_o \cos \phi$ [$I_o \Rightarrow$ No load current]
 $\cos \phi =$ Power factor]

Now $P_c = V_p I_o \cos \phi$

$\Rightarrow I_c = \frac{P_c}{V_p} = \frac{100}{230} = 0.435 \text{ A}$ (2)

b) $I_o = \frac{\text{No load 'VA'}}{V_p} = \frac{400}{230} = 1.739 \text{ A}$

$\therefore \cos \phi = \frac{I_c}{I_o} = \frac{0.435}{1.739} = 0.25$

$\therefore$ Magnetizing component of No-load current (2)

$I_m = I_o \sin \phi = 1.739 \times \sqrt{1 - (0.25)^2} = 1.684 \text{ A}$

c) Total No load current $I_0 = \frac{400}{230} = 1.739 \text{ A.}$

$\phi = \cos^{-1}(0.25) = 75.52^\circ.$ (2)

$\therefore$ No load current $\bar{I}_0 = \frac{1.739}{-75.52^\circ}$ [Here supply voltage is the reference].

d) No load power factor $\cos\phi = \frac{P_0}{S_{\text{no-load}}}$ (2)

$\Rightarrow \cos\phi = \frac{100}{400} = 0.25$ (lagging)

e) Rated Full load current of secondary winding $I_2 = \frac{\text{Full load 'VA'}}{V_S}$

$= \frac{5 \times 10^3}{115} = 43.478 \text{ A}$ (2)

4) 3 ϕ , 460V (Line-Line), 60Hz, 4 Pole induction motor has a slip of 5%.

a) The speed of rotating magnetic field is the synchronous speed of the motor (N_s).

where $N_s = \frac{120 \times f}{P}$ ~~$\frac{120 \times 60}{4}$~~

So for the given conditions —

$N_s = \frac{120 \times 60}{4} = 1800 \text{ rpm.}$ (2)

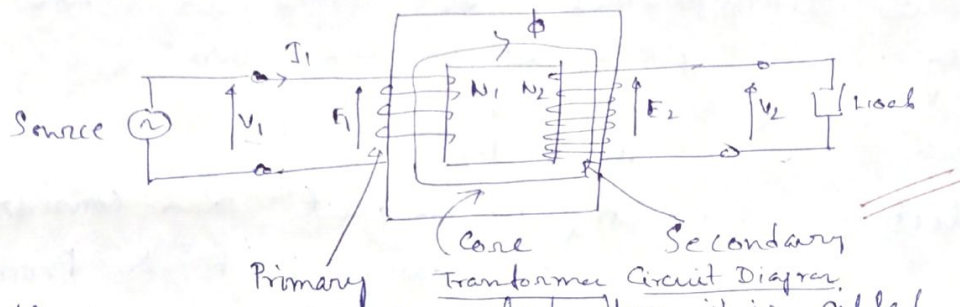
b) Frequency of rotor current (f_r) will be at the slip frequency.

$\therefore f_r = s \times f = 0.05 \times 60 = 3 \text{ Hz.}$ (2)

ii)

~~A transformer is made~~

A single phase transformer, in general, is made up of a core and two sets of windings. The windings are wound on the core. Generally the winding connected to an ac source is known as primary winding and the winding connected to the load is known as secondary winding. The transformer is used to facilitate energy transfer from source to load where the source and load ~~voltage~~ operating voltages are at different levels. The transformer can step-up or step-down the supplied ac voltage.



If Load is not connected then it is called No-load operation. If the transformer is connected to a ac voltage source at the Primary side and no load is connected at the secondary then only a small component of current is drawn at the primary side. This component of current, known as magnetizing current, is responsible for establishing a sinusoidal flux in the core. The sinusoidal flux induces voltage on primary & secondary winding. In an ideal situation all the flux produced by primary winding links the secondary winding and stays within the core.

Now if load is connected on the secondary side then the current flowing in the secondary winding creates a de-magnetizing effect in the core. This creates a tendency for reduction in voltage induced in the secondary windings. This in turn draws additional current in the primary side. The additional current compensates the de-magnetizing effect created by secondary winding. Thus the original flux level is maintained but ~~even though~~ although the primary & secondary windings are electrically isolated, while secondary side is loaded then primary side current also increases.

In a transformer the volt/turn for primary and secondary are same.

$$\text{Hence } \frac{V_1}{N_1} = \frac{V_2}{N_2} \quad \dots (1)$$

Neglecting magnetizing current and considering MMF balance the secondary MMF & Primary MMF are same.

$$\text{Hence } N_2 I_2 = N_1 I_1 \quad \dots (2)$$

Using eqn. (1) & (2) it can be written that $V_1 I_1 = V_2 I_2$, which means the ideal transformer works as an energy transferring device.

(3)

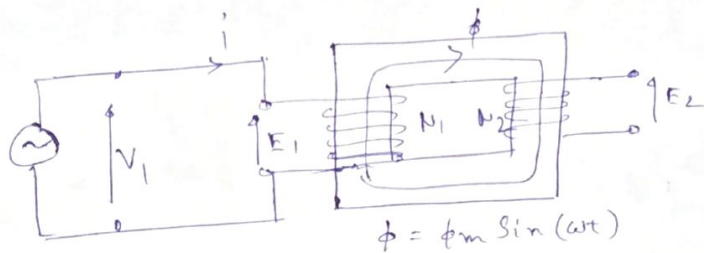
When a sinusoidal voltage is applied to a transformer primary winding then a small sinusoidal current is drawn to establish the flux in the core. The flux is also sinusoidal. Assume the flux ' ϕ ' is given by —

$$\phi = \phi_m \sin(\omega t)$$

where ϕ_m = Maximum value of flux in the core

$\omega \Rightarrow$ frequency in rad/s

$$\& \omega = 2\pi f \quad [f \Rightarrow \text{Supply frequency}]$$



Transformer Under No load.

$\therefore$ From Len's law induced e.m.f. in the transformer primary winding —

$$E_1 = -N_1 \frac{d\phi}{dt} \quad [N_1 \Rightarrow \text{Number of turns in primary winding}]$$

$$\Rightarrow E_1 = -N_1 \frac{d}{dt} (\phi_m \sin \omega t)$$

$$\Rightarrow E_1 = -N_1 \phi_m \omega \cos(\omega t) = N_1 \phi_m \omega \sin(\omega t - \frac{\pi}{2})$$

RMS value of E_1 will be —

$$E_1 = \frac{N_1 \phi_m \omega}{\sqrt{2}} = \sqrt{2} \pi N_1 \phi_m f$$

$$\Rightarrow E_1 = 4.44 N_1 \phi_m f$$

RMS value of secondary induced emf will be —
 $E_2 = \frac{N_2}{N_1} E_1 = \frac{N_2}{N_1} 4.44 N_1 \phi_m f$

(2)